Fiscal Year:	FY 2004	**Task Last Updated:**	FY 01/21/2005
PI Name:	Ewert, Michael K Ph.D.		
Project Title:	Advanced Life Support Systems Integration Modeling and Analysis		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Advanced life support		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	michael.k.ewert@nasa.gov	**Fax:**	FY 281-483-9167
PI Organization Type:	NASA CENTER	**Phone:**	281-483-9134
Organization Name:	NASA Johnson Space Center		
PI Address 1:	Mail Code EC2		
PI Address 2:	2101 NASA Road One		
PI Web Page:			
City:	Houston	**State:**	TX
Zip Code:	77586	**Congressional District:**	22
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	INTERNAL PROJECT
Start Date:	09/01/1998	**End Date:**	09/01/2006
No. of Post Docs:	0	**No. of PhD Degrees:**	2
No. of PhD Candidates:	2	**No. of Master' Degrees:**	6
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: End date changed to 9/1/2006, from 9/1/2016, because project no longer part of "human research project"--is now Exploration Technology Development Program and continues through that program (per D. Barta/JSC)--6/2008		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	None		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	SIMA will develop, apply and maintain methods of systems analysis and engineering for space vehicles and surface habitat life support systems to guide investments in research, resolve and integrate competing requirements and guide evolution of ALS architecture. SIMA will support all Advanced Life Support Elements with subsystem and system level analysis and maintain math models and reference documents necessary to accomplish that objective. One particular metric that will be computed annually is the ALS R&TD Metric which measures technology development and system architecture progress. Systems analysis is necessary to guide the ALS project in its technology development and testing efforts. Also, the ALS R&TD Metric must be reported per the Government Performance and Results Act.		

Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Systems Integration Modeling and Analysis (SIMA) supports the Advanced Life Support (ALS) project elements and management with guiding analyses to ensure sound invesments in technology development. These technologies often have terrestrial applications such as reducing the impact on Earth's eco-systems or energy savings.
Task Progress:	ALS analysis activities were coordinated across the agency and with several universities and grant holders. Monthly telecons and various technical reports were the main avenues used to accomplish this. Major SIMA products for FY04 included an annual update of the ALS Research and Technology Development Metric, a major revision of the ALS Baseline Values and Assumptions Document, version 4.0 of the ALS Sizing Analysis Tool and several system level technology trade studies.
Bibliography Type:	Description: (Last Updated: 06/03/2008)
Papers from Meeting Proceedings	Yeh HY, Brown CB, Jeng FF, Lin CH, Ewert MK. "ALSSAT Development Status and Its Applications in Trade Studies." International Conference on Environmental Systems, Colorado Springs, CO, July 2004. International Conference on Environmental Systems paper # 2004-01-2438, July 2004. , Jul-2004